Data Collection

Futures Rolls, Corporate Actions, and Point-in-Time Data

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 2

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A model needs one price series per asset. Exchanges quote prices per **instrument**, and instruments change: a futures contract expires, a stock splits, a company spins off a division or is acquired. The series has to be constructed from the instruments, and the construction embeds choices that the model then treats as data.

The two asset classes in this handout break in different ways:

	Commodity futures	Equities
What breaks	The instrument *ends*	The instrument *changes*
Frequency	Every 1–3 months, by design	Irregular, event-driven
Fix	Splice contracts, adjust the seam	Rescale the history at each event
Failure mode	Fake returns at roll dates	Fake returns at ex-dates
Second-order	Which contract? Rolled when?	What was known, and when?

The invariant. An adjusted price series is not a series of prices anyone traded. Its **levels** are an artifact of the adjustment convention; only its **returns** are economically meaningful. Never regress on, threshold on, or report the level of an adjusted series without knowing which convention produced it.

1 Commodity Futures

1.1 Contract Lifecycle and Expiration

A futures contract is a standardised agreement to buy or sell a fixed quantity and grade of a commodity at a set price on a set delivery date. Standardisation is what makes it fungible and liquid; the fixed delivery date is what makes it useless as a long-run price series. WTI crude alone lists monthly contracts years out, but any single one

— say CLH24, the March 2024 contract — trades for a bounded window and then stops.

Feature	Consequence for data
Fixed contract size (1,000 bbl WTI; 5,000 bu wheat)	Volume is comparable across time
Named delivery month and location	Prices are not interchangeable across months
Finite life (typically 1–2 years listed)	No single contract spans a backtest
Daily mark-to-market	Settlement price, not close, is the official mark

Contracts are named by a root symbol, a month code, and a year:

F (Jan)	G (Feb)	H (Mar)	J (Apr)	K (May)	M (Jun)
N (Jul)	Q (Aug)	U (Sep)	V (Oct)	X (Nov)	Z (Dec)

Not every root lists every month. Agricultural contracts follow the crop cycle, and the listed months encode it — which is exactly the information a continuous series destroys.

Sector	Root	Contract	Listed months
Energy	CL	Crude Oil WTI (NYMEX)	All
	NG	Natural Gas (NYMEX)	All
	HO	Heating Oil (NYMEX)	All
	RB	RBOB Gasoline (NYMEX)	All
	BZ	Brent Crude (NYMEX/ICE)	All
Agriculture	ZC	Corn (CBOT)	H, K, N, U, Z
	ZS	Soybeans (CBOT)	F, H, K, N, Q, U, X
	ZW	SRW Wheat (CBOT)	H, K, N, U, Z
	ZM	Soybean Meal (CBOT)	F, H, K, N, Q, U, V, Z
	ZL	Soybean Oil (CBOT)	F, H, K, N, Q, U, V, Z
	KE	HRW Wheat (CBOT)	H, K, N, U, Z
Softs	CT	Cotton (ICE)	H, K, N, V, Z
	CC	Cocoa (ICE)	H, K, N, U, Z
	KC	Coffee (ICE)	H, K, N, U, Z
	SB	Sugar #11 (ICE)	H, K, N, V
	OJ	Orange Juice (ICE)	F, H, K, N, U, X
Metals	GC	Gold (COMEX)	G, J, M, Q, V, Z
	SI	Silver (COMEX)	H, K, N, U, Z
	HG	Copper (COMEX)	H, K, N, U, Z
	PL	Platinum (NYMEX)	F, J, N, V
	PA	Palladium (NYMEX)	H, M, U, Z

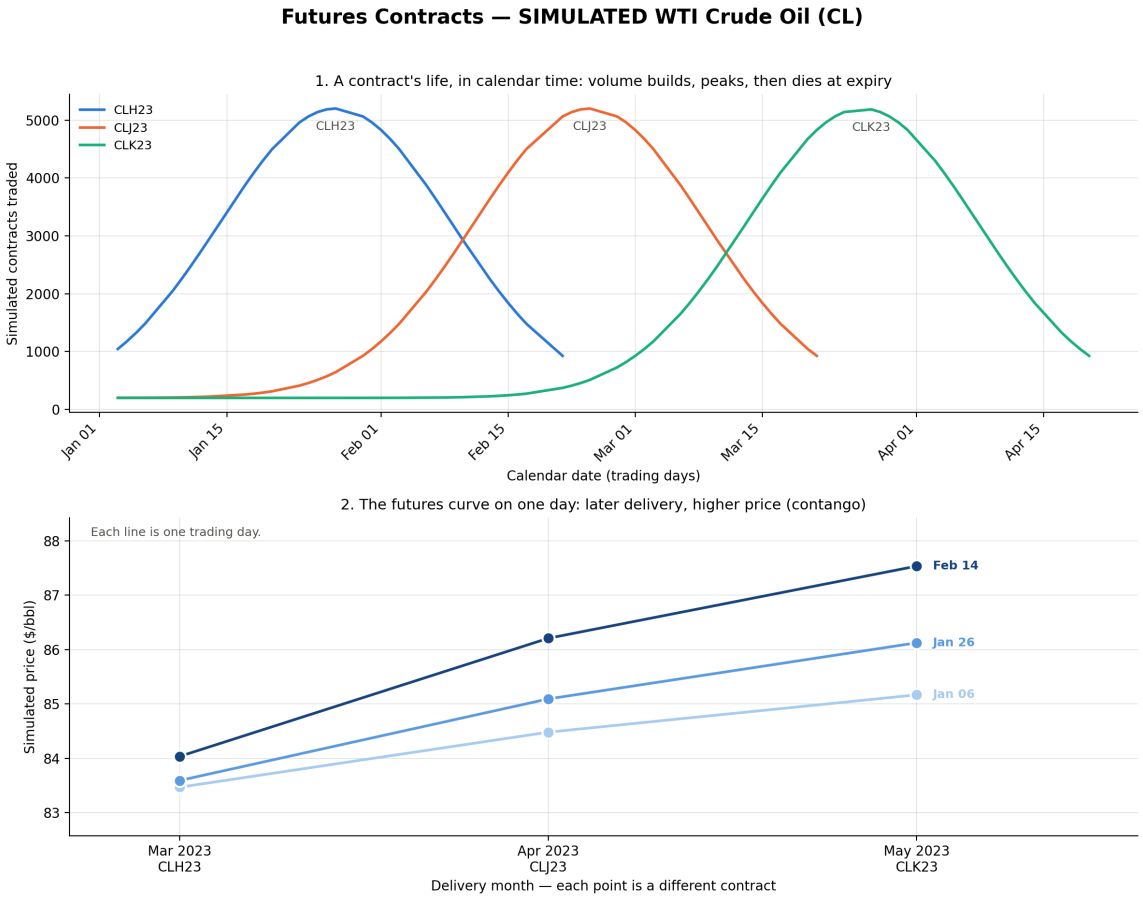
1.2 Continuous Contracts

A **continuous** (or *generic*) contract is a synthetic series built by splicing individual contracts end to end. Notation is positional: CL1 is the nearest listed contract at each date, CL2 the second nearest, and so on. CL1 is not an instrument — on any given day it *refers* to one, and the reference changes at roll dates.

Construction requires four decisions. They are independent of one another, and any combination is valid:

1. **Roll trigger** — the rule that fixes the first day of the roll.
2. **Roll window** — the number of days n over which the position moves from the old contract to the new one. $n = 1$ is a single-day switch.
3. **Weight schedule** — for $n > 1$, how much of the position moves on each day of the window.
4. **Price adjustment** — how historical levels are restated so the splice does not manufacture a return.

Decisions 1–3 determine *which contracts are held* on each date. Decision 4 determines *what number is recorded* for history. Whatever is chosen must be applied identically across the whole history.



Top: each contract's volume builds, peaks, and dies — liquidity migrates from CLH23 to CLJ23 to CLK23 in an overlapping relay, so "the" front month is a choice, not a fact. **Bottom:** on any single day the curve across delivery months is upward-sloping (contango here); the level gap between adjacent contracts is precisely the gap a roll must handle. Simulated data.

1.3 Roll Rules

Roll trigger

Rule	Trigger	Advantage	Cost
Time-based	Fixed offset, e.g. n business days before last trade date	Deterministic; computable in advance; no lookahead	Can roll into or out of thin liquidity
Volume-based	Roll when Vol(next) > Vol(front)	Tracks where trading actually is	Timing unknown ex ante; needs daily data; can oscillate
Open-interest	Roll when OI(next) > OI(front)	Reflects positioning, not day-trading; stable	OI is published with a lag; coarser signal

Open interest is the number of contracts currently outstanding — one long and one short, unclosed. Unlike volume it is a *stock*, not a flow: it rises only when a new long and a new short are created, and falls when a position is offset. It therefore measures committed capital rather than turnover, which is why it makes a steadier roll trigger.

Example (time-based, WTI). CLH24 last trades 20 February 2024. A "5 business days" rule rolls on 13 February, from CLH24 to CLJ24. The date is known months ahead, which is what makes the rule safe inside a backtest.

A volume rule can leak lookahead. "Roll on the first day the next contract's volume exceeds the front's" is decidable only *after* that day's volume prints. If the backtest switches contracts using that day's own close, it has used information the trader did not have. Roll on the following open, or use the previous day's volumes as the trigger.

Roll window and weight schedule

With a window of n days starting at τ_0, let w_t be the fraction of the position still in the old contract A at the close of day t, falling from 1 to 0 across the window. The held position on each day is

$$P_t = w_t P_t^A + (1 - w_t) P_t^B, \qquad t \in [\tau_0,\, \tau_0 + n - 1].$$

Common schedules:

- **Single switch** ($n = 1$): w goes from 1 to 0 in one day. Bloomberg CL1 and most vendor generics.
- **Linear**: $w_t = (\tau_1 - t)/n$, an equal slice each day. UNG rolls over four days; the S&P GSCI over five days at 20% per day.
- **Volume-weighted**: the daily slice is proportional to that day's volume in the new contract, so the move follows liquidity.

A window spreads the roll across several days, mirroring how a large position is actually executed and removing sensitivity to any single day's spread. It also blurs the term structure during the window, so it is a poor choice when the object of study *is* the term structure. A window does not remove the phantom return at the roll; it distributes it across n days. The adjustment in §1.4 is still required.

1.4 Price Adjustment

Adjacent contracts trade at different prices — the spread reflects storage cost, convenience yield, financing, and differing supply/demand expectations across delivery periods, not any change in the commodity's value. Splicing raw prices therefore inserts a return equal to the calendar spread on every roll. That return is fictitious. Over a ten-year backtest with monthly rolls it is roughly 120 fictitious returns.

Method	Definition	Use when
None (raw)	$P_t^{\text{cont}} = P_t^{\text{active}}$. Levels are real traded prices; jumps at rolls.	You need actual price levels: margin, contract specs, reporting. *Never* for returns.
Additive (back-adjustment)	Shift all history by the roll-date difference so the seam is flat.	Return and volatility work in a fixed-dollar contract; the default.
Multiplicative (ratio adjustment)	Scale all history by the roll-date price ratio.	Percentage-based work; guarantees positivity.

Additive

Rolling from contract A to contract B on date τ, with $\delta = P_\tau^B - P_\tau^A$:

$$P_t^{\text{adj}} = \begin{cases} P_t^{\text{raw}} + \delta, & t < \tau, \\ P_t^{\text{raw}}, & t \geq \tau. \end{cases}$$

Adjustments accumulate backwards through every prior roll, so the shift applied to a date ten years ago is the sum of all intervening δ's. In a persistently contangoed market that sum is large and negative, and **the adjusted series can go below zero.** Differences are preserved, which is all the method promises, but percentage returns $P_t/P_{t-1} - 1$ are meaningless on an additively adjusted series. Use price *differences* divided by a contemporaneous price level, or use multiplicative adjustment.

Multiplicative

With $\rho = P_\tau^B / P_\tau^A$:

$$P_t^{\text{adj}} = \begin{cases} P_t^{\text{raw}} \times \rho, & t < \tau, \\ P_t^{\text{raw}}, & t \geq \tau. \end{cases}$$

Ratios compound multiplicatively, prices stay positive, and percentage returns are correct across the seam. Dollar differences are distorted: a \$1 move ten years ago does not correspond to \$1 of P&L on one contract.

Adjustment inside a roll window

With $n > 1$ the roll is n partial rolls, and each day of the window carries its own adjustment. The raw blended series changes by

$$P_t - P_{t-1} = \underbrace{w_{t-1}\,(P_t^A - P_{t-1}^A) + (1 - w_{t-1})\,(P_t^B - P_{t-1}^B)}_{\text{P\&L of the position held overnight}} + \underbrace{(w_{t-1} - w_t)\,(P_t^B - P_t^A)}_{\text{phantom}}.$$

The phantom term is that day's slice times that day's spread. The additive adjustment on day t is therefore

$$\delta_t = (w_{t-1} - w_t)\,(P_t^B - P_t^A),$$

applied to all prices before t, exactly as for a single-day roll. Summed over the window the total shift is the slice-weighted average spread rather than the spread on one day.

For the multiplicative version, let $\tilde{P}_t = w_{t-1}P_t^A + (1 - w_{t-1})P_t^B$ be the position held overnight valued at today's prices. Then

$$\frac{P_t}{P_{t-1}} = \underbrace{\frac{\tilde{P}_t}{P_{t-1}}}_{\text{gross return on the position held overnight}} \times \underbrace{\frac{P_t}{\tilde{P}_t}}_{\text{phantom}},$$

so the ratio adjustment on day t is

$$\rho_t = \frac{P_t}{\tilde{P}_t} = \frac{w_t P_t^A + (1 - w_t)P_t^B}{w_{t-1}P_t^A + (1 - w_{t-1})P_t^B} = 1 + \frac{\delta_t}{\tilde{P}_t},$$

applied to all prices before t. With $n = 1$ this reduces to $\rho = P_\tau^B / P_\tau^A$.

Worked example

On 13 February 2024, CLH24 settles at \$75.50 and CLJ24 at \$76.20. A historical CLH24 observation on 15 January was \$73.80.

Method	Factor	Adjusted 15 Jan price
Additive	$\delta = 76.20 - 75.50 = +0.70$	$73.80 + 0.70 = \$74.50$
Multiplicative	$\rho = 76.20/75.50 = 1.00927$	$73.80 \times 1.00927 = \$74.48$
None	—	\$73.80, with a +0.93% fake return at the roll

The two adjusted answers differ by two cents here. Over 120 rolls and a wide price range they diverge substantially — which is why the convention must be recorded alongside the data.

Because the adjustment is anchored to the current contract, every roll restates the entire history: the adjusted price for a date ten years ago changes each month. Store the raw contract prices and the roll schedule (dates and weights), and rebuild the adjusted series on demand.

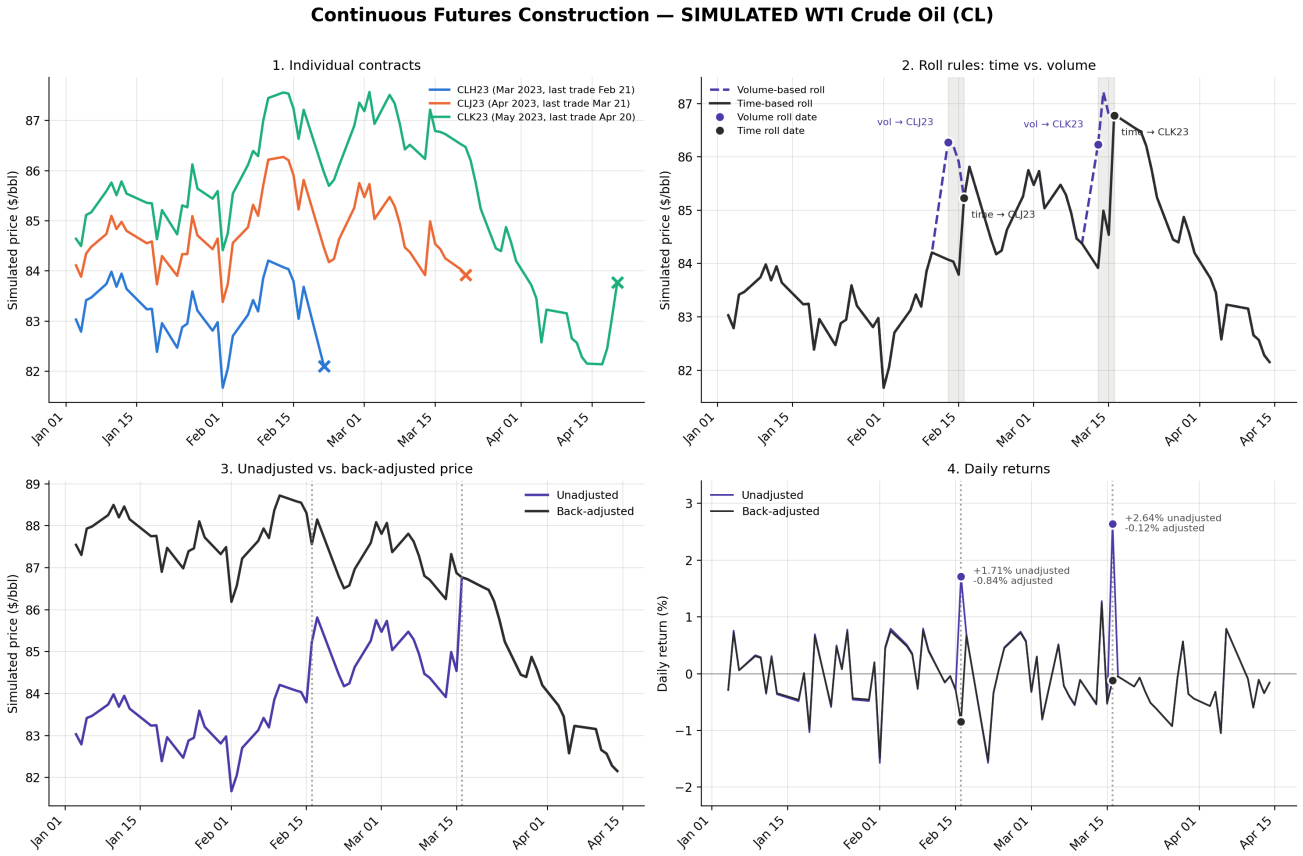
1.5 Behaviour Near Expiration

Contracts become badly behaved in their final days:

1. **Convergence.** The futures price is pulled to spot, so its dynamics stop being those of a forward-looking asset.
2. **Delivery pressure.** Participants unable or unwilling to make or take physical delivery must exit regardless of price.
3. **Thin liquidity.** Open interest has already migrated to the next contract; the same order moves the price much further.
4. **Settlement mechanics.** Contracts settling against a window average or an index become mechanically pinned or gamed near the fix.

Natural gas is the standard illustration: daily volatility of roughly 3–5% ten days out can run 8–12% three days out, with intraday moves above 20% on the final trading day. Feeding those observations into a GARCH fit or a VaR model contaminates the estimate with mechanics that have nothing to do with the commodity. Twenty percent of a contract's observations can easily be from its illiquid tail.

Practical rule. Roll 5–10 trading days before last trade date, and drop the final week when estimating volatility. Filter on volume and open interest rather than on days-to- expiry alone — the thresholds that matter are liquidity thresholds. The 2020 negative WTI settlement is the extreme case: a front-month contract whose holders had no delivery capacity, days from expiry, in a market with no storage left.



The construction as a four-step process. **Step 1** — the raw material: three overlapping contracts, each ending at its last trade date. **Step 2** — choose the roll trigger: time-based and volume-based rules roll on different days and produce different series. Here the window is one day, so decisions 2 and 3 are fixed. **Step 3** — choose the adjustment: unadjusted vs. additive; the adjusted series sits at a different level entirely, and the level is meaningless. **Step 4** — the result: the unadjusted series shows +1.71% and +2.64% returns on the roll dates where the true returns were −0.84% and −0.12%. Simulated data.

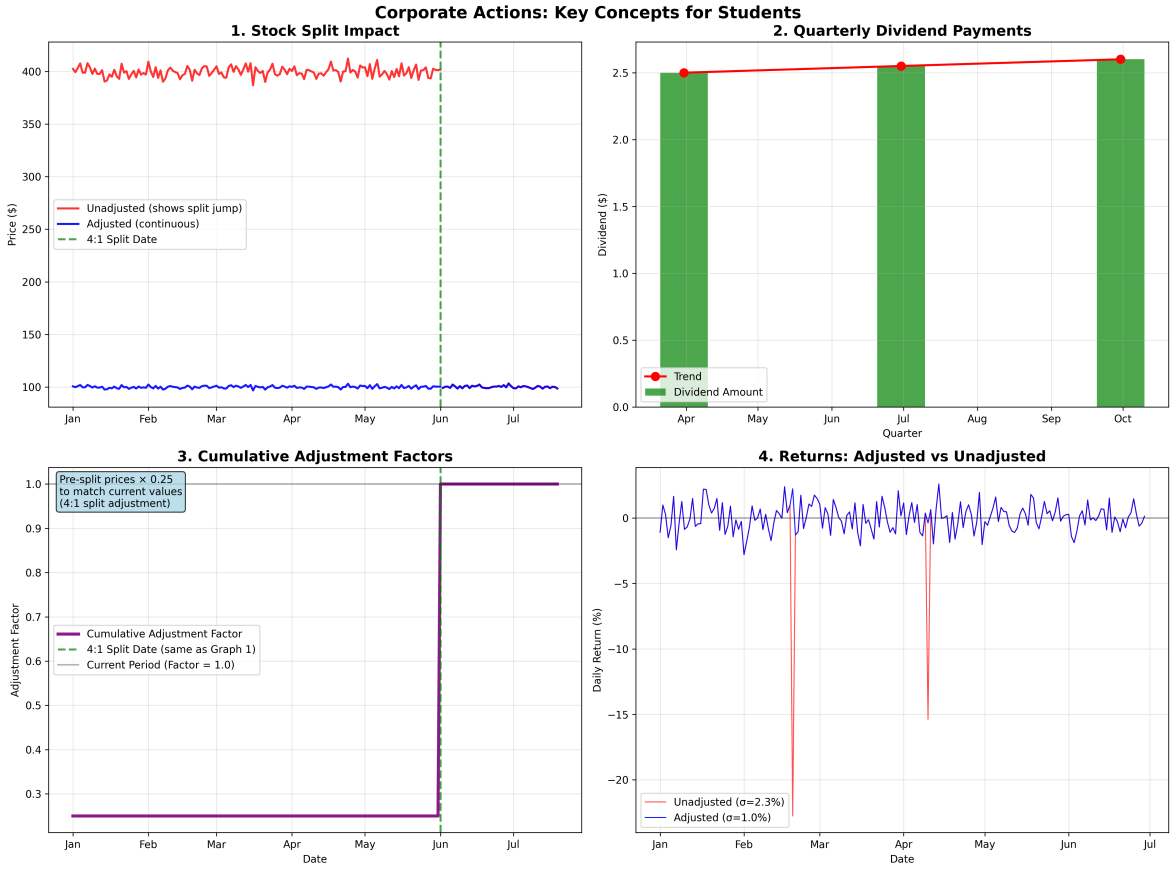
2 Equities: Corporate Actions

An equity does not expire, but its share count, price, and sometimes its identity change by corporate decision. A **corporate action** is any such event. The mandatory ones — splits, dividends, spin-offs — happen to every holder automatically; the voluntary ones — rights offerings, tender offers, elective stock dividends — require a choice, and the choice is itself part of the return.

Unadjusted equity prices are not wrong; they are simply prices of a share whose definition changed. A 4:1 split turns one share into four, and the price falls by three-quarters. Nothing happened to anyone's wealth, but a naive return calculation records −75%.

2.1 The Events

Action	What changes	Adjustment
Stock split / reverse split	Share count $\times n$, price $\div n$	Exact, known ratio; no ambiguity
Stock dividend	Shares issued pro rata	Treated as a fractional split
Cash dividend	Price drops $\approx$ dividend on ex-date	Additive or multiplicative factor
Special dividend	Large, irregular cash distribution	Same, but the size makes errors visible
Spin-off	Holder receives shares in a new entity	Requires valuing the distributed stub
Rights offering	Right to buy below market	Theoretical ex-rights price
Merger / acquisition	Security ceases or is exchanged	Series ends or is chained to the acquirer
Buyback	Share count falls, no per-share event	*No* price adjustment



(1) A 4:1 split in the raw series is a cliff; the adjusted series is continuous. **(3)** The cumulative adjustment factor is a step function, equal to 1 in the current period by construction and 0.25 before the split. **(4)** The consequence for a model: unadjusted daily returns carry two spurious spikes and report $\sigma = 2.3\%$ against a true 1.0%. Simulated data.

2.2 The Mathematics of Adjustment

Direction: backward is the standard

Backward adjustment rescales history so that the most recent price is the actual traded price. **Forward adjustment** holds history fixed and rescales the present. For an n:1 split on date t_s:

$$\text{backward:}\quad P_t^{\text{adj}} = \begin{cases} P_t^{\text{raw}}/n, & t < t_s \\ P_t^{\text{raw}}, & t \geq t_s \end{cases} \qquad \text{forward:}\quad P_t^{\text{adj}} = \begin{cases} P_t^{\text{raw}}, & t < t_s \\ P_t^{\text{raw}} \times n, & t \geq t_s \end{cases}$$

Backward adjustment is the convention in academic research and in essentially every vendor's "adjusted close", because the series then ends at recognisable, current prices. Its cost is that the series is **revised**: today's split changes yesterday's stored value, so an adjusted price database is not append-only, and a backtest run last month is not reproducible from this month's snapshot unless the adjustment history is versioned.

Apple, 31 August 2020, 4:1. Close on 28 August was \$499.23; the split-adjusted value stored today is \$124.81. The first post-split close was \$129.04. The one-day return is $129.04/124.81 - 1 = +3.4\%$, not $129.04/499.23 - 1 = -74\%$. Both numbers appear in the wild, depending on which column was read.

Dividends

A cash dividend of D with ex-date t_d removes D of value per share. Additively, $P_t^{\text{adj}} = P_t^{\text{raw}} - D$ for $t < t_d$ — which, applied across decades of dividends, drives long histories negative for the same reason back-adjusted futures go negative. Vendors therefore use the multiplicative form:

$$f = \frac{P_{t_d-1}^{\text{raw}} - D}{P_{t_d-1}^{\text{raw}}}, \qquad P_t^{\text{adj}} = P_t^{\text{raw}} \times f \quad \text{for } t < t_d.$$

The factor is the fraction of value that remains with the share. It is strictly between 0 and 1, so prices stay positive, and it composes cleanly with split factors.

The cumulative factor

In practice a vendor stores one **cumulative adjustment factor** F_t per date, the product of every split and dividend factor between t and today:

$$F_t = \prod_{e\,:\,t_e > t} f_e, \qquad P_t^{\text{adj}} = P_t^{\text{raw}} \times F_t, \qquad F_{\text{today}} = 1.$$

Volume must be adjusted by the reciprocal of the *split* component only — a split multiplies share count, a dividend does not:

$$V_t^{\text{adj}} = V_t^{\text{raw}} \,/\, S_t, \qquad S_t = \prod_{e \,\in\, \text{splits},\, t_e > t} \frac{1}{n_e}.$$

Applying the full price factor to volume is one of the most common silent errors in constructed datasets: dollar volume ceases to be consistent and any liquidity filter built on it drifts.

2.3 Spin-offs and Rights Offerings

Splits and dividends have known, exact factors. These two do not.

Spin-off (eBay → PayPal, ex-date Monday 20 July 2015). Each EBAY share received one PYPL share. On Friday 17 July, the last day with PayPal attached, EBAY closed at \$66.29. On the ex-date EBAY opened at \$26.89 and closed at \$28.57; PYPL opened at \$41.46 and closed at \$40.47. The adjustment factor for EBAY history is the fraction of the combined value that remained in the parent, but that fraction can only be measured at some moment τ after the separation, and it is a market outcome, not a contractual ratio. CRSP uses the ex-date closes, $28.57/(28.57 + 40.47) = 0.4138$ (`facpr` $= 1.4165$); the opening prints give $26.89/(26.89 + 41.46) = 0.3934$.

The choice of τ amounts to assuming the PYPL shares are sold at τ and the proceeds reinvested in EBAY. The factor rescales every pre-spin price by the same constant, so it changes only the ex-date return: the combined EBAY+PYPL return from Friday's close to τ, followed by EBAY alone from τ to Monday's close. That is +4.15% with the close factor, all of it the combined position's, and +9.55% with the open factor (+3.11% overnight, then EBAY's own +6.25% on the day). The earlier τ, the more of the ex-date return belongs to the parent alone; the ideal is the when-issued prices on 17 July, which value both pieces at the same moment as the \$66.29 cum-distribution close. Vendors disagree — when-issued, opening print, VWAP, close — because the earlier prices are cleaner in principle but thinner and noisier in practice, and CRSP's daily opens only begin in 1992.

Rights offering. With one right per five shares held, each right buying one new share at \$20, against a \$25 market price, the theoretical ex-rights price is

$$\text{TERP} = \frac{(5 \times 25) + (1 \times 20)}{5 + 1} = \frac{145}{6} = \$24.17,$$

so the factor is $24.17/25 = 0.9668$. But this is the value only to a holder who *exercises*. A holder who lets the rights lapse is genuinely diluted, and the adjusted series — which assumes exercise — overstates their return. The adjusted series encodes a policy, not just an arithmetic identity.

2.4 Total Return

Price adjustment for dividends reconstructs the total-return series implicitly: the factor assumes the dividend is reinvested at the ex-date price. Stated directly,

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1, \qquad \text{TRI}_t = \text{TRI}_{t-1}(1 + R_t), \qquad \text{TRI}_0 = 100.$$

Do not do both. Computing $(P_t^{\text{adj}} + D_t)/P_{t-1}^{\text{adj}} - 1$ on a dividend-adjusted series double-counts the dividend. Either use adjusted prices and simple returns, or use unadjusted prices and add the cash flow explicitly. Deciding which is a matter of reading the vendor's column definitions, not of preference.

2.5 Vendor Differences

Every provider makes the same set of choices and they do not make them identically.

Vendor	Dividend threshold	Method	Spin-off treatment
Bloomberg	Configurable (often \$0.10)	Multiplicative	Full adjustment
LSEG/Refinitiv	Configurable (often \$0.05)	Multiplicative	Full adjustment
Yahoo Finance	None	Multiplicative	Partial / inconsistent
CRSP	None	Additive & multiplicative factors published	Full adjustment

The differences that actually bite are: whether small dividends are adjusted at all; whether the announcement or the ex-date drives the factor; how spin-off stubs are valued; and how delistings, bankruptcies, and foreign actions are treated. CRSP publishes the raw factors separately from the adjusted price, which is why it is the reference for academic work — you can reconstruct any convention from it. Most commercial feeds ship the adjusted price only.

Survivorship bias. A universe assembled from currently listed tickers has already deleted every company that failed, merged, or delisted. Backtested returns on such a universe are biased upward, often by 1–3% annually. This is not a corporate-actions bug — it is a *universe construction* bug, and no amount of price adjustment fixes it. Use a point-in-time universe with delisting returns attached.

3 Point-in-Time Data and the Knowledge Date

Everything above is about getting the numbers right. This section is about getting the *timing* of the numbers right, and it is the failure mode that quietly ruins backtests.

Three dates, routinely conflated:

- **Knowledge date** — when the information became public (announcement).
- **Effective date** — when the event takes effect in the market (ex-date).
- **As-of date** — the date a value is *about*, which may be revised later.

A **point-in-time** database answers "what did the world look like on date t, using only what was known by t?" — which requires storing both the as-of date and the knowledge date for every value.

Look-ahead bias is using information that was not yet available. It is rarely deliberate and almost never obvious, because the adjusted price series does not carry a timestamp saying when it acquired its current shape.

Tesla, 2022. The split announcement and the split itself are two weeks apart:

Date	Event
10 August 2022	Nothing known. Unadjusted prices are what a trader saw.
11 August 2022	3:1 split announced — **knowledge date**.
17 August 2022	Record date.
25 August 2022	Split effective — **ex-date**. Adjusted prices apply from here.

A strategy that trades on 12 August knows the split is coming, but the shares have not split; the price is still ∼\$900. If the backtest reads a split-adjusted \$300 on that date and sizes a position from it, share counts and dollar exposure are wrong by a factor of three — even though the returns are correct. Returns and levels fail differently, and only levels fail here, which is why the bug survives casual checking.

The problem is much worse in fundamentals than in prices. Earnings are restated; index membership is revised; a value "as of" Q2 was not known until the filing weeks later. Using the restated figure at the original as-of date is look-ahead bias in a form that no price adjustment can detect.

Practice.

1. Store as-of *and* knowledge dates for every value; never overwrite in place.
2. Drive decisions off the knowledge date; drive price adjustment off the effective date.
3. Version the corporate-actions table — an adjusted price series is a *view*, and the factors are the underlying data.
4. Test for lookahead directly: re-run a backtest against a snapshot rebuilt as of an earlier date and confirm the signals match.
5. Record the vendor, the extraction date, and the adjustment convention with every dataset.

4 Cross-Asset Consistency

4.1 Timestamps

Multi-asset work fails on timestamps before it fails on anything else. A "daily close" is a different instant in every market:

Instrument	Close	Note
US equities (NYSE/Nasdaq)	16:00 ET	Official close; auction print
E-mini S&P 500 (CME)	16:00 ET (settlement)	Nearly 24h session; settlement $\neq$ last trade
WTI Crude (NYMEX)	14:30 ET (settlement)	Settlement is a window average
Brent Crude (ICE)	19:30 London	Different day boundary from NYMEX
Gold (COMEX)	13:30 ET (settlement)	

Correlating an equity close with a crude close aligns 16:00 ET against 14:30 ET: 90 minutes of equity moves have no crude counterpart, and the measured correlation is attenuated. Across Asian and European sessions the mismatch can be a full day, which silently converts a contemporaneous relationship into a lead-lag one.

1. Store everything in UTC with an explicit timezone; convert only for display.
2. Prefer the *settlement* price to the last trade for futures — it is the official mark and the basis for margin.
3. Handle DST transitions explicitly; US and European changeover dates differ by weeks.
4. For intraday work, resample every series onto a common grid before joining, and be explicit about whether the bar is left- or right-labelled.

4.2 Holiday Calendars

Venues close on different days, so a multi-asset panel is ragged by construction:

Holiday	US Equities	NYMEX	ICE	LME
New Year's Day	Closed	Closed	Closed	Closed
Good Friday	Closed	Closed	Closed	Closed
Independence Day	Closed	Closed	Open	Open
Thanksgiving	Closed	Closed	Open	Open
Christmas Day	Closed	Closed	Closed	Closed
Boxing Day	Open	Open	Closed	Closed

The handling choice changes the statistics:

Strategy	Effect	Distortion
Forward fill	Carry last price forward	Injects zero returns; deflates volatility, inflates autocorrelation
Interpolate	Estimate from neighbours	Fabricates data; never acceptable for returns
Drop the date	Use only dates all venues traded	Safest; costs observations; a two-day return is labelled one day
Union of dates	Keep all, mark missing as `NaN`	Honest; forces the model to handle missingness

Dropping non-common dates is the usual default for daily cross-asset work. Whichever is used, the gap must remain visible: a return computed over a three-day weekend is not a one-day return, and annualising by $\sqrt{252}$ without accounting for it is wrong.

5 Validation

Constructed series should be checked mechanically, because the errors are systematic rather than random and are therefore invisible in summary statistics.

```
def validate_adjustments(prices, actions, threshold=0.05):
    """Flag returns above `threshold` with no corporate action on that date."""
    returns = prices.pct_change()
    suspects = returns[returns.abs() > threshold]
    return [
        (date, r) for date, r in suspects.items()
        if date not in actions.index
    ]
```

A useful minimum set of checks:

1. **Unexplained jumps.** Every return beyond a threshold should map to a news event, a corporate action, or a roll date. Unmapped jumps are data errors until proven otherwise.
2. **Roll-date returns.** On an adjusted futures series, roll-date returns should be statistically indistinguishable from other days. If they are systematically large, the adjustment is not being applied.
3. **Price/volume consistency.** Split factors applied to price must appear as the reciprocal in volume. Check

dollar volume for step changes at split dates.

4. **Cross-vendor reconciliation.** Compare adjusted returns across two sources; differences localise precisely to the events they disagree about.
5. **Reconstruction.** Rebuild the adjusted series from raw prices and the factor table and confirm it matches what is stored. If it cannot be rebuilt, the convention is not documented well enough.
6. **Negative and zero prices.** Legitimate in a back-adjusted futures series; almost always an error in an equity series.

6 Summary

Concept	What it means
Continuous contract	Synthetic series splicing individual futures; CL1, CL2, ...
Roll rule	When the reference contract changes: time, volume, or open interest
Back-adjustment	Additive seam fix; preserves differences; can go negative
Ratio adjustment	Multiplicative seam fix; preserves returns; stays positive
Corporate action	Any issuer event changing share count, price, or identity
Cumulative factor F_t	Product of all future split and dividend factors; $F_{\text{today}} = 1$
Knowledge date	When information became public
Effective date	When the event took effect in the market
Look-ahead bias	Using information not yet available at the decision time
Survivorship bias	Universe built from survivors; unfixable by price adjustment

The four rules.

1. **Levels are conventions; returns are data.** Know which series you hold before computing anything on it.
2. **One convention per dataset, applied to the whole history.** Mixed conventions produce artifacts that look like signal.
3. **Store factors, not just adjusted prices.** Adjusted prices are a view; factors plus raw prices are the data, and only they permit reconstruction.
4. **Every value carries two dates.** What it is about, and when it became known. A backtest that does not distinguish them is measuring the future.

7 References

- Geman, H. *Commodities and Commodity Derivatives.* Wiley — contract specifications, term structure, storage and convenience yield.
- Harris, L. *Trading and Exchanges.* Oxford — market mechanics, settlement, and where prices come from.
- CRSP, *Calculations and Index Methodologies* — the reference treatment of adjustment factors, delisting returns, and point-in-time construction.
- Bailey, Borwein, López de Prado, Zhu, "Pseudo-Mathematics and Financial Charlatanism," *Notices of the AMS* (2014) — what backtest bias costs.
- CME Group and ICE contract specifications — the authoritative source for expiry, settlement, and delivery for each root symbol.

Source	Strength	Typical use
CRSP	Survivorship-bias-free; factors published	Academic research, backtesting
Bloomberg	Breadth, real-time, cross-asset	Trading, research
LSEG/Refinitiv	Corporate actions, fundamentals	Risk, compliance
Nasdaq Data Link	APIs, alternative datasets	Quantitative research
Norgate, CSI	Futures continuous-contract construction	Systematic futures backtesting